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Question 1

Correct

Mark 0.25 out of
0.25

Which service-to-host model allows multiple different services to be deployed together on the same host in fixed combinations?

- ☐ a. Single Service Instance per Host
- ☒ b. Multiple Static Services per Host ✓
- ☐ c. Monolithic Deployment
- ☐ d. Multiple Scheduled Services per Host

**The correct answer is:****Multiple Static Services per Host**

Question 2

Correct

Mark 0.25 out of
0.25

In microservices, what is the main benefit of having separate CI/CD pipelines for each service?

- ☐ a. It increases build time for all services
- ☐ b. It forces services to share the same deployment cycle
- ☐ c. It limits scalability of the architecture
- ☒ d. It reduces dependency and allows independent deployment ✓

The correct answer is:**It reduces dependency and allows independent deployment**

Question 3

Correct

Mark 0.25 out of
0.25

When building a microservice in Django, what is the main purpose of using Django REST Framework (DRF)?

- ☐ a. To design HTML templates
- ☐ b. To handle background jobs only
- ☐ c. To manage static files
- ☒ d. To enable RESTful API creation for services ✓

The correct answer is: To enable RESTful API creation for services

Question 4

Correct

Mark 0.25 out of
0.25

What is the main purpose of running multiple instances of a microservice?

- ☒ a. To improve availability and handle higher loads ✓
- ☐ b. To simplify debugging
- ☐ c. To reduce memory usage
- ☐ d. To reduce code duplication



The correct answer is:
To improve availability and handle higher loads

Question 5

Correct

Mark 0.25 out of
0.25

What does automated rollback in CI/CD pipelines do?

- ☐ a. Deletes old containers permanently
- ☐ b. Manually requests approval for rollback
- ☐ c. Runs database migrations automatically
- ☒ d. Restores a previous stable version when a deployment fails ✓

The correct answer is:
Restores a previous stable version when a deployment fails

Question 6

Correct

Mark 0.25 out of
0.25

What is the purpose of the Bulkhead pattern in microservices?

- ☒ a. To isolate service components so that a failure in one doesn't affect others ✓
- ☐ b. To increase the number of network calls
- ☐ c. To ensure all requests are processed by a single instance
- ☐ d. To merge multiple APIs into one

The correct answer is:

To isolate service components so that a failure in one doesn't affect others

Question 7

Correct

Mark 0.25 out of
0.25

In a Django-based microservice, which component is primarily responsible for handling incoming HTTP requests and returning responses?

- ☐ a. Middleware
- ☒ b. Views ✓
- ☐ c. Models
- ☐ d. Templates



The correct answer is: Views

Question 8

Correct

Mark 0.25 out of
0.25

When using event sourcing, transactions are managed by:

- ☒ a. Storing all events that change the state ✓
- ☐ b. Storing only the current state of data
- ☐ c. Sharing a global transaction log
- ☐ d. Using two-phase commit

The correct answer is: Storing all events that change the state

Question 9

Correct

Mark 0.25 out of
0.25

Which of the following is a drawback of using an API Gateway?

- ☐ a. Encapsulates internal structure
- ☒ b. Becomes a development bottleneck ✓
- ☐ c. Provides authentication
- ☐ d. Simplifies client code

The correct answer is: Becomes a development bottleneck

Question 10

Correct

Mark 0.25 out of
0.25

What does graceful degradation mean in microservice design?

- ☐ a. Shutting down all services when one fails
- ☐ b. Reducing system capacity manually
- ☐ c. Restarting all microservices periodically
- ☒ d. Maintaining limited functionality instead of complete failure during issues ✓

The correct answer is: Maintaining limited functionality instead of complete failure during issues

Question 11

Correct

Mark 0.25 out of
0.25

In DevOps tooling, which tool is commonly used to define infrastructure as code (IaC) within automated pipelines?

- ☒ a. Terraform ✓
- ☐ b. Grafana
- ☐ c. Docker
- ☐ d. Postman

The correct answer is:
Terraform

Question 12

Correct

Mark 0.25 out of 0.25

In the “Single Service Instance per Host” model, each host (VM or container) runs:

- ☐ a. Multiple unrelated services
- ☒ b. A single instance of one service ✓
- ☐ c. A random number of services
- ☐ d. A load balancer and multiple services

The correct answer is: A single instance of one service

Question 13

Correct

Mark 0.25 out of 0.25

Which of the following statements about pipeline automation is TRUE?

- ☐ a. Automated pipelines cannot handle parallel builds
- ☐ b. Automation eliminates the need for source control
- ☐ c. Manual intervention is required for every build
- ☒ d. Automation increases consistency and reduces human error ✓

The correct answer is: Automation increases consistency and reduces human error

Question 14

Correct

Mark 0.25 out of 0.25

What is the main advantage of event-driven communication between microservices?

- ☐ a. Tight coupling of services
- ☒ b. Loose coupling and better scalability ✓
- ☐ c. Direct data sharing between services
- ☐ d. Simplified debugging

The correct answer is:
Loose coupling and better scalability

Question 15

Correct

Mark 0.25 out of 0.25

What is the role of health probes (liveness and readiness) in microservices?

- ☐ a. To perform backups automatically
- ☒ b. To help orchestrators detect unhealthy instances and restart them ✓
- ☐ c. To increase latency intentionally
- ☐ d. To test database schema changes

The correct answer is:

To help orchestrators detect unhealthy instances and restart them

Question 16

Correct

Mark 0.25 out of 0.25

The Backends for Frontends (BFF) pattern is beneficial because it:

- ☐ a. Reduces observability
- ☒ b. Makes APIs independently scalable ✓
- ☐ c. Increases startup time
- ☐ d. Couples all APIs together



The correct answer is:

Makes APIs independently scalable

Question 17

Correct

Mark 0.25 out of 0.25

What is the primary goal of load balancing in microservices?

- ☒ a. To distribute incoming traffic evenly across multiple service instances ✓
- ☐ b. To merge multiple APIs into one
- ☐ c. To reduce the number of services in production
- ☐ d. To synchronize database replicas

The correct answer is:

To distribute incoming traffic evenly across multiple service instances

Question 18

Correct

Mark 0.25 out of
0.25

What is the safest way to manage secrets (like API keys) in a code repository?

- ☐ a. Email them to developers
- ☐ b. Store them in a .env file committed to Git
- ☒ c. Use a secrets manager or environment variables outside version control ✓
- ☐ d. Hardcode them into the source code

The correct answer is:

Use a secrets manager or environment variables outside version control

Question 19

Correct

Mark 0.25 out of
0.25

In securing a Git repository, which practice is most recommended?

- ☐ a. Disabling commit history
- ☐ b. Sharing SSH keys between all team members
- ☐ c. Pushing all branches to a public repo
- ☒ d. Using signed commits and branch protection rules ✓



The correct answer is: Using signed commits and branch protection rules

Question 20

Correct

Mark 0.25 out of
0.25

What does JWT (JSON Web Token) primarily help with in a service?

- ☒ a. Authenticating and securely transmitting user identity information ✓
- ☐ b. Encrypting database data
- ☐ c. Caching static assets
- ☐ d. Performing load balancing

The correct answer is:

Authenticating and securely transmitting user identity information

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